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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 1	
TITLE: Java Environment Setup and Basic Program Execution	

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyse outputs, and apply the concepts to solve practical programming problems.

Programme

Assignment1.java

```java
// Java Program to demonstrate Variables, Methods and Constructors

class Student {

    String name;            // Instance variables
    int age;

    Student() {               // Default Constructor
        name="Riya";
        age=19;
    }

    void display() {                // Method to display details
        System.out.println("Student Name : " +name);
        System.out.println("Student Age  : " +age);
    }

    int add(int a,int b) {            // Method to calculate sum
        return a+b;
    }
}

public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
        int x=20;               // Local variables
        int y=30;

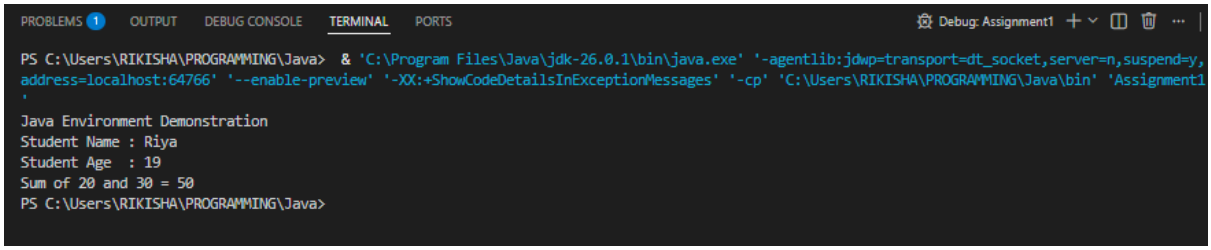
        // Creating object
        Student s=new Student();

        System.out.println(x: "Java Environment Demonstration");

        s.display();
        int result=s.add(x,y);

        System.out.println("Sum of " +x+ " and " +y+ " = " +result);
    }
}
```

Output



Conclusion

The program was executed successfully in the Java environment. It demonstrated the use of the JDK/JRE/JVM, variables, methods, constructors, object creation, and method invocation. The output verified that Java programs can effectively use object-oriented concepts to solve programming problems.

Questions

1. Explain the main concept used in Java environment setup, variables and methods.
   The Java environment consists of JDK, JRE, and JVM, which are used to develop, compile, and run Java programs. Variables are used to store data, while methods are blocks of code that perform specific tasks and improve program organization.
2. What are the advantages of using Java variables and methods?
   - Variables make data storage and manipulation easy.
   - Methods promote **code reusability** and reduce duplication.
   - They improve **readability** and simplify program maintenance.
   - Methods help organize code into smaller, manageable parts.
3. How can this implementation be improved?
   - By taking **user input** instead of using fixed values.
   - By using **parameterized constructors**.
   - By adding **input validation** and exception handling.
   - By applying better object-oriented principles such as encapsulation.
4. What are the real-world applications?
   Java variables and methods are used in:

- Banking and financial software
- E-commerce applications
- Mobile apps (Android)
- Desktop applications
- Enterprise and web-based systems
- Student and employee management system